

Table 2: Performance comparison of PostgreSQL and ClickHouse

Metric	PostgreSQL (Row)	ClickHouse (Columnar)
Data scanned	~200GB (all columns)	~20GB (2 columns)
Query time (cold)	~380 seconds	~1.2 seconds
Storage size	~150GB	~18GB (compressed)
Suitable workload	OLTP/point reads	OLAP/aggregations

These numbers are illustrative but consistent with publicly reported benchmarks from ClickHouse, Snowflake, and Redshift engineering blogs. Actual results vary by hardware, indexing, and query complexity.

Architectural patterns and best practices

The Lambda architecture: A common enterprise pattern separates a row-based operational store from a columnar analytical store. Change Data Capture (CDC) tools such as Debezium stream row-level changes from PostgreSQL or MySQL into a columnar warehouse (Redshift, Snowflake, BigQuery) in near real time. This decouples OLTP and OLAP workloads cleanly.

Materialised views for pre-aggregation: In columnar databases, materialised views can pre-compute expensive aggregations and refresh them on a schedule. ClickHouse Materialized Views use an INSERT trigger pattern:

```
-- Materialised view: pre-aggregate daily revenue
CREATE MATERIALIZED VIEW daily_revenue_mv
ENGINE = SummingMergeTree()
ORDER BY (day, region)
AS
SELECT
    toDate(created_at) AS day,
    region,
    sum(quantity * unit_price) AS revenue
FROM orders
GROUP BY day, region;
```

Partitioning strategies

Columnar databases gain additional speed from partition pruning. Partition by a high-cardinality time or category column to skip irrelevant data files entirely.

- **Time partitioning:** Partition by year/month for time-series data (BigQuery, Redshift).
- **Range partitioning:** Split by numeric ranges for customer segments.
- **Hash partitioning:** Distribute data evenly across nodes when no natural ordering exists.

The open source ecosystem offers mature, production-grade options for both storage models. The tools given below are grouped by storage paradigm, covering relational databases, analytical engines, embedded stores, and replication/streaming layers.

Row-based (OLTP) open source tools

PostgreSQL: The world's most advanced open source relational database. PostgreSQL uses a heap-based row store with MVCC (Multi-Version Concurrency Control) for high-throughput OLTP. It supports JSON, full-text search, geospatial data via PostGIS, and logical replication — making it a near-universal choice for application backends.

Licence: PostgreSQL licence (Permissive)

URL: [postgresql.org](https://www.postgresql.org)

MySQL/MariaDB: MySQL remains the most deployed open source RDBMS, powering most LAMP-stack applications. MariaDB is its fully open source fork with additional storage engines (Aria, Spider, ColumnStore). Both use InnoDB as their default row-based storage engine, offering clustered primary-key indexes and row-level locking.

Licence: GPL v2

URL: [mysql.com/mariadb.org](https://www.mysql.com/mariadb.org)

SQLite: A serverless, self-contained, row-based SQL engine embedded in an application process. SQLite is the most widely deployed database in the world (mobile apps, browsers, IoT). It stores an entire database in a single cross-platform file. Ideal for local state, testing, and edge devices where network latency is unacceptable.

Licence: Public domain

URL: [sqlite.org](https://www.sqlite.org)

CockroachDB: A distributed, horizontally scalable SQL database modelled after Google Spanner. CockroachDB stores rows in a distributed key-value store (RocksDB), providing ACID transactions, geo-partitioning, and PostgreSQL wire-protocol compatibility. It auto-rebalances shards across nodes without downtime.

Licence: BSL 1.1 / CCL

URL: [cockroachlabs.com](https://www.cockroachlabs.com)

Columnar (OLAP) open source tools

ClickHouse: Originally built at Yandex for web analytics, ClickHouse is now the fastest open source columnar database for real-time analytical queries. Its MergeTree engine stores data in sorted, compressed column files. It supports vectorised SIMD execution, approximate aggregations, and materialised views. Commonly used for observability, product analytics, and ad-tech pipelines.

Licence: Apache 2.0

URL: clickhouse.com